Documentation

Fast D1 Documentation

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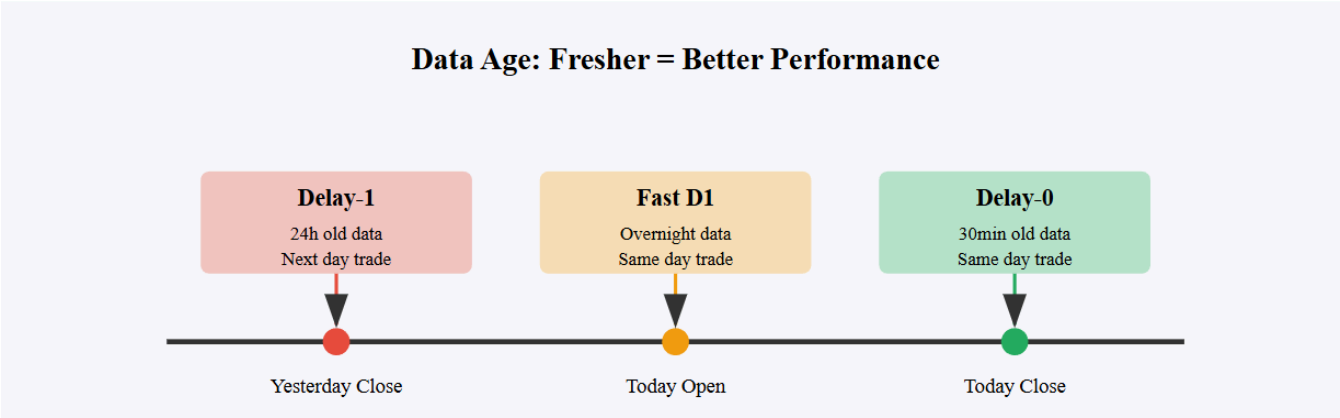
What is Fast D1?

Fast D1 is a framework for developing Alphas that use data available between day 0 (D0) and day 1 (D1). By leveraging the most current overnight information - such as news, earnings announcements, analyst updates, and pre-market activity - Fast D1 Alphas can capture predictive signals that deliver superior performance compared to regular D1 models.

Key Advantage: Access to real-time overnight catalysts that traditional D1 Alphas miss.



How Fast D1 Works

Data Timing Comparison

Understanding the critical difference in data availability:

Delay	Field Type	Data Timestamp	Information Captured
1	snt_buzz	Yesterday's market close	Yesterday's End-of-day data only
Fast D1	snt_buzz_fast_d1	Today's market open	Overnight news, events, pre-market moves
0	snt_buzz	30 minutes before today's close	Close to end of today – harder to execute

Fast D1 captures the overnight information gap that drives market movements at the open

Implementation Guide

How to Simulate Fast D1 Alphas

Simple 2-Step Process:

1. **Select Delay-1** in simulation settings
2. **Use fields with _fast_d1 suffix**

Important Notes

✅ **Every Fast D1 field has a regular D1 equivalent** (just without the **_fast_d1** suffix)
✅ **No separate delay dropdown** for Fast D1-it's the same Delay-1 setting
✅ **Currently available only in the USA region**
✅ **Same submission criteria** as regular D1 alphas

Example Field Naming Convention

Regular D1 Field: snt_buzz

Fast D1 Field: snt_buzz_fast_d1

news21	news	83
option24	option	82
other696	other	66
earningscall_sentiment	other	59
model240	model	52
model216	model	45
model136	model	45
other384	other	33
news46	news	30
earnings7	earnings	20
other296	other	19
news7	news	19
other492	other	15
news36	news	14
broker24	broker	13
search_interest	other	12
news104	news	11
news13	news	8
fundamental110	fundamental	7
earnings1	earnings	6
socialmedia12	socialmedia	6
pv90	pv	4
news59	news	2
news35	news	2
shortinterest30	shortinterest	2
other10	other	1
	Total	1049

Additionally, you must use the datafield '*pv98_vwap_15m_delay_minus_1_returns_fast_d1*' for using returns value and '*usa_vwap_15m_delay_minus_1_fast_d1'* for using the price of the stock at the FastD1 cutoff time.

fields using the Brain API using Python.

Best Practices & Technical Considerations

- Almost 50% of the Fast D1 fields available for use are vector datafields. So make sure to use appropriate vector operators for such fields to prevent errors during simulation.
- Try using the winsorize operator or other outlier removal techniques incase you face weight concentration errors.

Submission Guidelines

✅ Mandatory Performance Check

Submit Fast D1 Alphas ONLY if they outperform their D1 equivalents

Example:

Alpha A: Using snt_buzz_fast_d1 → Sharpe 2.8, Turnover 33%

Alpha B: Using snt_buzz (regular D1) → Sharpe 2.6, Turnover 30%

✓ Submit Alpha A only if performance is superior

Production Correlation

- **Same correlation pool** as regular D1 alphas
- **Approach:** Using delta (**field_fast_d1 - field**) may help reduce production correlation

Research Directions & Alpha Development

🚀 Priority 1: Overnight Information Capture

Focus on datasets that provide unique overnight insights:

High Priority:

- **News datasets** (news21, news46, news7, news36, news104)

- **Earnings data** (earnings7, earnings4, earnings8al_sentiment)
- **Short interest changes** (shortinterest30)

💡 **Priority 2: Delta/Change-Based Alphas**

Capture the **overnight information change**:

 # Example: Overnight social media sentiment shift : Change in snt_buzz_fast_d1 compared to snt_buzz

🎯 **Priority 3: Event-Driven Models**

Target specific overnight catalysts:

- Earnings announcements
- Analyst upgrades/downgrades
- News events
- Pre-market price/returns discovery
- Short interest information

Note: You can submit high turnover Alphas (~30%) on those Alphas that capture short lived overnight dislocations. However, ensure that you maintain a Minimum Margin > 4bps in these Alphas

Quick Start Checklist

- Set simulation to **Delay-1**
- Select fields with **_fast_d1** suffix
- Use **vector operators** for vector fields
- Use special **returns and price fields**
- Apply **winsorize** if needed
- **Compare performance** with regular D1 version
- Verify **margin > 4 bps**
- Test for **production correlation**
- Submit only if **Fast D1 version outperforms D1 – an exception to this is if you create Alphas using delta between Fast D1 field and D1 field as then you do not have comparable D1 performance.**

Key Takeaways

✨ **Fast D1 captures the overnight edge** that traditional D1 misses
✨ **Focus on news, options, social media, and earnings** datasets
✨ **Use delta approaches** (fast_d1 - regular) for unique signals
✨ **Submit only superior performers** compared to D1 equivalents